



Microsoft

Exam Questions DP-203

Data Engineering on Microsoft Azure

About ExamBible

Your Partner of IT Exam

Found in 1998

ExamBible is a company specialized on providing high quality IT exam practice study materials, especially Cisco CCNA, CCDA, CCNP, CCIE, Checkpoint CCSE, CompTIA A+, Network+ certification practice exams and so on. We guarantee that the candidates will not only pass any IT exam at the first attempt but also get profound understanding about the certificates they have got. There are so many alike companies in this industry, however, ExamBible has its unique advantages that other companies could not achieve.

Our Advances

* 99.9% Uptime

All examinations will be up to date.

* 24/7 Quality Support

We will provide service round the clock.

* 100% Pass Rate

Our guarantee that you will pass the exam.

* Unique Gurantee

If you do not pass the exam at the first time, we will not only arrange FULL REFUND for you, but also provide you another exam of your claim, ABSOLUTELY FREE!

NEW QUESTION 1

- (Exam Topic 1)

You need to design the partitions for the product sales transactions. The solution must mee the sales transaction dataset requirements.

What should you include in the solution? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point.

Answer Area

Partition product sales transactions
data by:

Sales date
Product ID
Promotion ID

Store product sales transactions data
in:

An Azure Synapse Analytics dedicated SQL pool
An Azure Synapse Analytics serverless SQL pool
An Azure Data Lake Storage Gen2 account linked to an Azure Synapse Analytics workspace

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

Box 1: Sales date

Scenario: Contoso requirements for data integration include:

➤ Partition data that contains sales transaction records. Partitions must be designed to provide efficient loads by month. Boundary values must belong to the partition on the right.

Box 2: An Azure Synapse Analytics Dedicated SQL pool Scenario: Contoso requirements for data integration include:

➤ Ensure that data storage costs and performance are predictable.

The size of a dedicated SQL pool (formerly SQL DW) is determined by Data Warehousing Units (DWU). Dedicated SQL pool (formerly SQL DW) stores data in relational tables with columnar storage. This format significantly reduces the data storage costs, and improves query performance.

Synapse analytics dedicated sql pool Reference:

<https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/sql-data-warehouse-overview-wha>

NEW QUESTION 2

- (Exam Topic 1)

You need to design a data storage structure for the product sales transactions. The solution must meet the sales transaction dataset requirements.

What should you include in the solution? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point.

Answer Area

Table type to store the product sales
transactions:

Hash
Round-robin
Replicated

When creating the table for sales transactions:

Configure a clustered index.
Set the distribution column to product ID.
Set the distribution column to the sales date.

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

Answer Area

Table type to store the product sales
transactions:

Hash
Round-robin
Replicated

When creating the table for sales transactions:

Configure a clustered index.
Set the distribution column to product ID.
Set the distribution column to the sales date.

NEW QUESTION 3

- (Exam Topic 3)

You have an Azure Synapse Analytics dedicated SQL pool.

You need to ensure that data in the pool is encrypted at rest. The solution must NOT require modifying applications that query the data.

What should you do?

- A. Enable encryption at rest for the Azure Data Lake Storage Gen2 account.
B. Enable Transparent Data Encryption (TDE) for the pool.
C. Use a customer-managed key to enable double encryption for the Azure Synapse workspace.
D. Create an Azure key vault in the Azure subscription grant access to the pool.

Answer: B

Explanation:

Transparent Data Encryption (TDE) helps protect against the threat of malicious activity by encrypting and decrypting your data at rest. When you encrypt your database, associated backups and transaction log files are encrypted without requiring any changes to your applications. TDE encrypts the storage of an entire database by using a symmetric key called the database encryption key.

Reference:

<https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/sql-data-warehouse-overviewmana>

NEW QUESTION 4

- (Exam Topic 3)

You have an on-premises data warehouse that includes the following fact tables. Both tables have the following columns: DateKey, ProductKey, RegionKey. There are 120 unique product keys and 65 unique region keys.

Table	Comments
Sales	The table is 600 GB in size. DateKey is used extensively in the WHERE clause in queries. ProductKey is used extensively in join operations. RegionKey is used for grouping. Severity-five percent of records relate to one of 40 regions.
Invoice	The table is 6 GB in size. DateKey and ProductKey are used extensively in the WHERE clause in queries. RegionKey is used for grouping.

Queries that use the data warehouse take a long time to complete.

You plan to migrate the solution to use Azure Synapse Analytics. You need to ensure that the Azure-based solution optimizes query performance and minimizes processing skew.

What should you recommend? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point

Table	Distribution type	Distribution column
Sales:	Hash-distributed Round-robin	DateKey ProductKey RegionKey
Invoices:	Hash-distributed Round-robin	DateKey ProductKey RegionKey

- A. Mastered
B. Not Mastered

Answer: A

Explanation:

Box 1: Hash-distributed

Box 2: ProductKey

ProductKey is used extensively in joins.

Hash-distributed tables improve query performance on large fact tables.

Box 3: Round-robin

Box 4: RegionKey

Round-robin tables are useful for improving loading speed.

Consider using the round-robin distribution for your table in the following scenarios:

- When getting started as a simple starting point since it is the default
- If there is no obvious joining key
- If there is not good candidate column for hash distributing the table
- If the table does not share a common join key with other tables
-

If the join is less significant than other joins in the query

➤ When the table is a temporary staging table

Note: A distributed table appears as a single table, but the rows are actually stored across 60 distributions. The rows are distributed with a hash or round-robin algorithm.

Reference:

<https://docs.microsoft.com/en-us/azure/sql-data-warehouse/sql-data-warehouse-tables-distribute>

NEW QUESTION 5

- (Exam Topic 3)

You are monitoring an Azure Stream Analytics job.

The Backlogged Input Events count has been 20 for the last hour. You need to reduce the Backlogged Input Events count.

What should you do?

- A. Drop late arriving events from the job.
- B. Add an Azure Storage account to the job.
- C. Increase the streaming units for the job.
- D. Stop the job.

Answer: C

Explanation:

General symptoms of the job hitting system resource limits include:

➤ If the backlog event metric keeps increasing, it's an indicator that the system resource is constrained (either because of output sink throttling, or high CPU).

Note: Backlogged Input Events: Number of input events that are backlogged. A non-zero value for this metric implies that your job isn't able to keep up with the number of incoming events. If this value is slowly increasing or consistently non-zero, you should scale out your job: adjust Streaming Units.

Reference:

<https://docs.microsoft.com/en-us/azure/stream-analytics/stream-analytics-scale-jobs> <https://docs.microsoft.com/en-us/azure/stream-analytics/stream-analytics-monitoring>

NEW QUESTION 6

- (Exam Topic 3)

You use Azure Data Factory to prepare data to be queried by Azure Synapse Analytics serverless SQL pools. Files are initially ingested into an Azure Data Lake Storage Gen2 account as 10 small JSON files. Each file contains the same data attributes and data from a subsidiary of your company.

You need to move the files to a different folder and transform the data to meet the following requirements: ➤ Provide the fastest possible query times.

➤ Automatically infer the schema from the underlying files.

How should you configure the Data Factory copy activity? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Copy behavior:

	▼
Flatten hierarchy	
Merge files	
Preserve hierarchy	

Sink file type:

	▼
CSV	
JSON	
Parquet	
TXT	

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

Box 1: Preserver herarchy

Compared to the flat namespace on Blob storage, the hierarchical namespace greatly improves the performance of directory management operations, which improves overall job performance.

Box 2: Parquet

Azure Data Factory parquet format is supported for Azure Data Lake Storage Gen2. Parquet supports the schema property.

Reference:

<https://docs.microsoft.com/en-us/azure/storage/blobs/data-lake-storage-introduction> <https://docs.microsoft.com/en-us/azure/data-factory/format-parquet>

NEW QUESTION 7

- (Exam Topic 3)

You need to schedule an Azure Data Factory pipeline to execute when a new file arrives in an Azure Data Lake Storage Gen2 container. Which type of trigger should you use?

- A. on-demand
- B. tumbling window
- C. schedule
- D. event

Answer: D

Explanation:

Event-driven architecture (EDA) is a common data integration pattern that involves production, detection, consumption, and reaction to events. Data integration scenarios often require Data Factory customers to trigger pipelines based on events happening in storage account, such as the arrival or deletion of a file in Azure Blob Storage account.

Reference:

<https://docs.microsoft.com/en-us/azure/data-factory/how-to-create-event-trigger>

NEW QUESTION 8

- (Exam Topic 3)

You have an Azure Data Lake Storage account that has a virtual network service endpoint configured.

You plan to use Azure Data Factory to extract data from the Data Lake Storage account. The data will then be loaded to a data warehouse in Azure Synapse Analytics by using PolyBase.

Which authentication method should you use to access Data Lake Storage?

- A. shared access key authentication
- B. managed identity authentication
- C. account key authentication
- D. service principal authentication

Answer: B

Explanation:

Reference:

<https://docs.microsoft.com/en-us/azure/data-factory/connector-azure-sql-data-warehouse#use-polybase-to-load-d>

NEW QUESTION 9

- (Exam Topic 3)

You are designing a statistical analysis solution that will use custom proprietary Python functions on near real-time data from Azure Event Hubs.

You need to recommend which Azure service to use to perform the statistical analysis. The solution must minimize latency.

What should you recommend?

- A. Azure Stream Analytics
- B. Azure SQL Database
- C. Azure Databricks
- D. Azure Synapse Analytics

Answer: A

NEW QUESTION 10

- (Exam Topic 3)

You are designing a real-time dashboard solution that will visualize streaming data from remote sensors that connect to the internet. The streaming data must be aggregated to show the average value of each 10-second interval. The data will be discarded after being displayed in the dashboard.

The solution will use Azure Stream Analytics and must meet the following requirements:

- Minimize latency from an Azure Event hub to the dashboard.
- Minimize the required storage.
- Minimize development effort.

What should you include in the solution? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point

Azure Stream Analytics input type:

	▼
Azure Event Hub	
Azure SQL Database	
Azure Stream Analytics	
Microsoft Power BI	

Azure Stream Analytics output type:

	▼
Azure Event Hub	
Azure SQL Database	
Azure Stream Analytics	
Microsoft Power BI	

Aggregation query location:

	▼
Azure Event Hub	
Azure SQL Database	
Azure Stream Analytics	
Microsoft Power BI	

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

Reference:
<https://docs.microsoft.com/en-us/azure/stream-analytics/stream-analytics-power-bi-dashboard>

NEW QUESTION 10

- (Exam Topic 3)

You plan to implement an Azure Data Lake Gen2 storage account.

You need to ensure that the data lake will remain available if a data center fails in the primary Azure region. The solution must minimize costs.

Which type of replication should you use for the storage account?

- A. geo-redundant storage (GRS)
- B. zone-redundant storage (ZRS)
- C. locally-redundant storage (LRS)
- D. geo-zone-redundant storage (GZRS)

Answer: A

Explanation:

Geo-redundant storage (GRS) copies your data synchronously three times within a single physical location in the primary region using LRS. It then copies your data asynchronously to a single physical location in the secondary region.

Reference:
<https://docs.microsoft.com/en-us/azure/storage/common/storage-redundancy>

NEW QUESTION 15

- (Exam Topic 3)

You have an Azure subscription that contains a logical Microsoft SQL server named Server1. Server1 hosts an Azure Synapse Analytics SQL dedicated pool named Pool1.

You need to recommend a Transparent Data Encryption (TDE) solution for Server1. The solution must meet the following requirements:

- Track the usage of encryption keys.
- Maintain the access of client apps to Pool1 in the event of an Azure datacenter outage that affects the availability of the encryption keys.

What should you include in the recommendation? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

To track encryption key usage:

▼

Always Encrypted
TDE with customer-managed keys
TDE with platform-managed keys

To maintain client app access in the event of a datacenter outage:

▼

Create and configure Azure key vaults in two Azure regions.
Enable Advanced Data Security on Server1.
Implement the client apps by using a Microsoft .NET Framework data provider.

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

Box 1: TDE with customer-managed keys

Customer-managed keys are stored in the Azure Key Vault. You can monitor how and when your key vaults are accessed, and by whom. You can do this by enabling logging for Azure Key Vault, which saves information in an Azure storage account that you provide.

Box 2: Create and configure Azure key vaults in two Azure regions

The contents of your key vault are replicated within the region and to a secondary region at least 150 miles away, but within the same geography to maintain high durability of your keys and secrets.

Reference:

<https://docs.microsoft.com/en-us/azure/synapse-analytics/security/workspaces-encryption> <https://docs.microsoft.com/en-us/azure/key-vault/general/logging>

NEW QUESTION 18

- (Exam Topic 3)

You are designing an Azure Databricks table. The table will ingest an average of 20 million streaming events per day.

You need to persist the events in the table for use in incremental load pipeline jobs in Azure Databricks. The solution must minimize storage costs and incremental load times.

What should you include in the solution?

- A. Partition by DateTime fields.
- B. Sink to Azure Queue storage.
- C. Include a watermark column.
- D. Use a JSON format for physical data storage.

Answer: B

Explanation:

The Databricks ABS-AQS connector uses Azure Queue Storage (AQS) to provide an optimized file source that lets you find new files written to an Azure Blob storage (ABS) container without repeatedly listing all of the files.

This provides two major advantages:

- Lower costs: no more costly LIST API requests made to ABS.

Reference:

<https://docs.microsoft.com/en-us/azure/databricks/spark/latest/structured-streaming/aqs>

NEW QUESTION 23

- (Exam Topic 3)

You have an Azure Synapse Analytics dedicated SQL pool that contains a large fact table. The table contains 50 columns and 5 billion rows and is a heap.

Most queries against the table aggregate values from approximately 100 million rows and return only two columns.

You discover that the queries against the fact table are very slow. Which type of index should you add to provide the fastest query times?

- A. nonclustered columnstore
- B. clustered columnstore
- C. nonclustered
- D. clustered

Answer: B

Explanation:

Clustered columnstore indexes are one of the most efficient ways you can store your data in dedicated SQL pool.

Columnstore tables won't benefit a query unless the table has more than 60 million rows. Reference:

<https://docs.microsoft.com/en-us/azure/synapse-analytics/sql/best-practices-dedicated-sql-pool>

NEW QUESTION 27

- (Exam Topic 3)

You have an Azure data factory.

You need to examine the pipeline failures from the last 60 days. What should you use?

- A. the Activity log blade for the Data Factory resource
- B. the Monitor & Manage app in Data Factory
- C. the Resource health blade for the Data Factory resource
- D. Azure Monitor

Answer: D

Explanation:

Data Factory stores pipeline-run data for only 45 days. Use Azure Monitor if you want to keep that data for a longer time.

Reference:

<https://docs.microsoft.com/en-us/azure/data-factory/monitor-using-azure-monitor>

NEW QUESTION 30

- (Exam Topic 3)

You need to implement an Azure Databricks cluster that automatically connects to Azure Data Lake Storage Gen2 by using Azure Active Directory (Azure AD) integration.

How should you configure the new cluster? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point.

Cluster Mode:

	▼
High Concurrency	
Premium	
Standard	

Advanced option to enable:

	▼
Azure Data Lake Storage Gen1 Credential Passthrough	
Table Access Control	

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

Box 1: High Concurrency

Enable Azure Data Lake Storage credential passthrough for a high-concurrency cluster. Incorrect:

Support for Azure Data Lake Storage credential passthrough on standard clusters is in Public Preview.

Standard clusters with credential passthrough are supported on Databricks Runtime 5.5 and above and are limited to a single user.

Box 2: Azure Data Lake Storage Gen1 Credential Passthrough

You can authenticate automatically to Azure Data Lake Storage Gen1 and Azure Data Lake Storage Gen2 from Azure Databricks clusters using the same Azure Active Directory (Azure AD) identity that you use to log into Azure Databricks. When you enable your cluster for Azure Data Lake Storage credential passthrough, commands that you run on that cluster can read and write data in Azure Data Lake Storage without requiring you to configure service principal credentials for access to storage.

References:

<https://docs.azuredatabricks.net/spark/latest/data-sources/azure/adls-passthrough.html>

NEW QUESTION 34

- (Exam Topic 3)

You have an Azure data factory.

You need to ensure that pipeline-run data is retained for 120 days. The solution must ensure that you can query the data by using the Kusto query language.

Which four actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.

NOTE: More than one order of answer choices is correct. You will receive credit for any of the correct orders you select.

Actions	Answer Area
Select the PipelineRuns category.	
Create a Log Analytics workspace that has Data Retention set to 120 days.	
Stream to an Azure event hub.	
Create an Azure Storage account that has a lifecycle policy.	
From the Azure portal, add a diagnostic setting.	
Send the data to a Log Analytics workspace.	
Select the TriggerRuns category.	

- A. Mastered
B. Not Mastered

Answer: A

Explanation:

Step 1: Create an Azure Storage account that has a lifecycle policy

To automate common data management tasks, Microsoft created a solution based on Azure Data Factory. The service, Data Lifecycle Management, makes frequently accessed data available and archives or purges other data according to retention policies. Teams across the company use the service to reduce storage costs, improve app performance, and comply with data retention policies.

Step 2: Create a Log Analytics workspace that has Data Retention set to 120 days.

Data Factory stores pipeline-run data for only 45 days. Use Azure Monitor if you want to keep that data for a longer time. With Monitor, you can route diagnostic logs for analysis to multiple different targets, such as a Storage Account: Save your diagnostic logs to a storage account for auditing or manual inspection. You can use the diagnostic settings to specify the retention time in days.

Step 3: From Azure Portal, add a diagnostic setting. Step 4: Send the data to a log Analytics workspace,

Event Hub: A pipeline that transfers events from services to Azure Data Explorer. Keeping Azure Data Factory metrics and pipeline-run data.

Configure diagnostic settings and workspace.

Create or add diagnostic settings for your data factory.

- In the portal, go to Monitor. Select Settings > Diagnostic settings.
- Select the data factory for which you want to set a diagnostic setting.
- If no settings exist on the selected data factory, you're prompted to create a setting. Select Turn on diagnostics.
- Give your setting a name, select Send to Log Analytics, and then select a workspace from Log Analytics Workspace.
- Select Save. Reference:

<https://docs.microsoft.com/en-us/azure/data-factory/monitor-using-azure-monitor>

NEW QUESTION 39

- (Exam Topic 3)

You have an Azure Factory instance named DF1 that contains a pipeline named PL1.PL1 includes a tumbling window trigger.

You create five clones of PL1. You configure each clone pipeline to use a different data source.

You need to ensure that the execution schedules of the clone pipeline match the execution schedule of PL1. What should you do?

- A. Add a new trigger to each cloned pipeline
B. Associate each cloned pipeline to an existing trigger.
C. Create a tumbling window trigger dependency for the trigger of PL1.
D. Modify the Concurrency setting of each pipeline.

Answer: B

NEW QUESTION 43

- (Exam Topic 3)

You are designing an inventory updates table in an Azure Synapse Analytics dedicated SQL pool. The table will have a clustered columnstore index and will include the following columns:

- EventDate: 1 million per day
 - EventTypeID: 10 million per event type
 - WarehouseID: 100 million per warehouse
 - ProductCategoryTypeID: 25 million per product category type
- You identify the following usage patterns:

Analyst will most commonly analyze transactions for a warehouse.

Queries will summarize by product category type, date, and/or inventory event type. You need to recommend a partition strategy for the table to minimize query times. On which column should you recommend partitioning the table?

- A. ProductCategoryTypeID
- B. EventDate
- C. WarehouseID
- D. EventTypeID

Answer: D

NEW QUESTION 48

- (Exam Topic 3)

You are creating dimensions for a data warehouse in an Azure Synapse Analytics dedicated SQL pool. You create a table by using the Transact-SQL statement shown in the following exhibit.

```
CREATE TABLE [DBO].[DimProduct] (  
    [ProductKey] [int] IDENTITY(1,1) NOT NULL,  
    [ProductSourceID] [int] NOT NULL,  
    [ProductName] [nvarchar](100) NOT NULL,  
    [ProductNumber] [nvarchar](25) NOT NULL,  
    [Color] [nvarchar](15) NULL,  
    [Size] [nvarchar](5) NULL,  
    [Weight] [decimal](8, 2) NULL,  
    [ProductCategory] [nvarchar](100) NULL,  
    [SellStartDate] [date] NOT NULL,  
    [SellEndDate] [date] NULL,  
    [RowInsertedDateTime] [datetime] NOT NULL,  
    [RowUpdatedDateTime] [datetime] NOT NULL,  
    [ETLAuditID] [int] NOT NULL  
)
```

Use the drop-down menus to select the answer choice that completes each statement based on the information presented in the graphic.
NOTE: Each correct selection is worth one point.

DimProduct is a **[answer choice]** slowly changing
dimension (SCD).

▼

Type 0

Type 1

Type 2

The ProductKey column is **[answer choice]**.

▼

a surrogate key

a business key

an audit column

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

Box 1: Type 2

A Type 2 SCD supports versioning of dimension members. Often the source system doesn't store versions, so the data warehouse load process detects and manages changes in a dimension table. In this case, the dimension table must use a surrogate key to provide a unique reference to a version of the dimension member. It also includes columns that define the date range validity of the version (for example, StartDate and EndDate) and possibly a flag column (for example, IsCurrent) to easily filter by current dimension members.

Reference:

<https://docs.microsoft.com/en-us/learn/modules/populate-slowly-changing-dimensions-azure-synapse-analytics>

NEW QUESTION 50

- (Exam Topic 3)

You are designing a fact table named FactPurchase in an Azure Synapse Analytics dedicated SQL pool. The table contains purchases from suppliers for a retail store. FactPurchase will contain the following columns.

Name	Data type	Nullable
PurchaseKey	Bigint	No
DateKey	Int	No
SupplierKey	Int	No
StockItemKey	Int	No
PurchaseOrderID	Int	Yes
OrderedQuantity	Int	No
OrderedOuters	Int	No
ReceivedOuters	Int	No
Package	Nvarchar(50)	No
IsOrderFinalized	Bit	No
LineageKey	Int	No

FactPurchase will have 1 million rows of data added daily and will contain three years of data. Transact-SQL queries similar to the following query will be executed daily.

```
SELECT
SupplierKey, StockItemKey, COUNT(*)
FROM FactPurchase
WHERE DateKey >= 20210101
AND DateKey <= 20210131
GROUP By SupplierKey, StockItemKey
Which table distribution will minimize query times?
```

- A. round-robin
- B. replicated
- C. hash-distributed on DateKey
- D. hash-distributed on PurchaseKey

Answer: D

Explanation:

Hash-distributed tables improve query performance on large fact tables, and are the focus of this article. Round-robin tables are useful for improving loading speed.

Reference:

<https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/sql-data-warehouse-tables-distribu>

NEW QUESTION 55

- (Exam Topic 3)

You need to output files from Azure Data Factory.

Which file format should you use for each type of output? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Columnar format:

▼

Avro

GZip

Parquet

TXT

JSON with a timestamp:

▼

Avro

GZip

Parquet

TXT

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

Box 1: Parquet

Parquet stores data in columns, while Avro stores data in a row-based format. By their very nature, column-oriented data stores are optimized for read-heavy analytical workloads, while row-based databases are best for write-heavy transactional workloads.

Box 2: Avro

An Avro schema is created using JSON format.

AVRO supports timestamps.

Note: Azure Data Factory supports the following file formats (not GZip or TXT).

- Avro format
- Binary format
- Delimited text format
- Excel format
- JSON format
- ORC format
- Parquet format
- XML format

Reference:

<https://www.datanami.com/2018/05/16/big-data-file-formats-demystified>

NEW QUESTION 58

- (Exam Topic 3)

Which Azure Data Factory components should you recommend using together to import the daily inventory data from the SQL server to Azure Data Lake Storage?

To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Integration runtime type:	Azure integration runtime Azure-SSIS integration runtime Self-hosted integration runtime
Trigger type:	Event-based trigger Schedule trigger Tumbling window trigger
Activity type:	Copy activity Lookup activity Stored procedure activity

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

Answer Area

Integration runtime type:	Azure integration runtime Azure-SSIS integration runtime Self-hosted integration runtime
Trigger type:	Event-based trigger Schedule trigger Tumbling window trigger
Activity type:	Copy activity Lookup activity Stored procedure activity

NEW QUESTION 61

- (Exam Topic 3)

You use Azure Data Lake Storage Gen2.

You need to ensure that workloads can use filter predicates and column projections to filter data at the time the data is read from disk.

Which two actions should you perform? Each correct answer presents part of the solution. NOTE: Each correct selection is worth one point.

- A. Reregister the Microsoft Data Lake Store resource provider.
- B. Reregister the Azure Storage resource provider.
- C. Create a storage policy that is scoped to a container.

- D. Register the query acceleration feature.
- E. Create a storage policy that is scoped to a container prefix filter.

Answer: BD

NEW QUESTION 66

- (Exam Topic 3)

You build an Azure Data Factory pipeline to move data from an Azure Data Lake Storage Gen2 container to a database in an Azure Synapse Analytics dedicated SQL pool.

Data in the container is stored in the following folder structure.

/in/{YYYY}/{MM}/{DD}/{HH}/{mm}

The earliest folder is /in/2021/01/01/00/00. The latest folder is /in/2021/01/15/01/45. You need to configure a pipeline trigger to meet the following requirements:

- Existing data must be loaded.
- Data must be loaded every 30 minutes.
- Late-arriving data of up to two minutes must be included in the load for the time at which the data should have arrived.

How should you configure the pipeline trigger? To answer, select the appropriate options in the answer area. NOTE: Each correct selection is worth one point.

Type:

Event
On-demand
Schedule
Tumbling window

Additional properties:

Prefix: /in/, Event: Blob created
Recurrence: 30 minutes, Start time: 2021-01-01T00:00
Recurrence: 30 minutes, Start time: 2021-01-01T00:00, Delay: 2 minutes
Recurrence: 32 minutes, Start time: 2021-01-15T01:45

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

Box 1: Tumbling window

To be able to use the Delay parameter we select Tumbling window. Box 2:

Recurrence: 30 minutes, not 32 minutes

Delay: 2 minutes.

The amount of time to delay the start of data processing for the window. The pipeline run is started after the expected execution time plus the amount of delay. The delay defines how long the trigger waits past the due time before triggering a new run. The delay doesn't alter the window startTime.

Reference:

<https://docs.microsoft.com/en-us/azure/data-factory/how-to-create-tumbling-window-trigger>

NEW QUESTION 71

- (Exam Topic 3)

You have an Azure Stream Analytics job that receives clickstream data from an Azure event hub.

You need to define a query in the Stream Analytics job. The query must meet the following requirements: ➤ Count the number of clicks within each 10-second window based on the country of a visitor.

- Ensure that each click is NOT counted more than once. How should you define the Query?

- A. SELECT Country, Avg(*) AS AverageFROM ClickStream TIMESTAMP BY CreatedAt GROUP BY Country, SlidingWindow(second, 10)
- B. SELECT Country, Count(*) AS CountFROM ClickStream TIMESTAMP BY CreatedAt GROUP BY Country, TumblingWindow(second, 10)
- C. SELECT Country, Avg(*) AS AverageFROM ClickStream TIMESTAMP BY CreatedAt GROUP BY Country, HoppingWindow(second, 10, 2)
- D. SELECT Country, Count(*) AS CountFROM ClickStream TIMESTAMP BY CreatedAt GROUP BY Country, SessionWindow(second, 5, 10)

Answer: B

Explanation:

Tumbling window functions are used to segment a data stream into distinct time segments and perform a function against them, such as the example below. The key differentiators of a Tumbling window are that they repeat, do not overlap, and an event cannot belong to more than one tumbling window.

Example: Reference:

<https://docs.microsoft.com/en-us/azure/stream-analytics/stream-analytics-window-functions>

NEW QUESTION 74

- (Exam Topic 3)

You have an enterprise data warehouse in Azure Synapse Analytics named DW1 on a server named Server1. You need to verify whether the size of the transaction log file for each distribution of DW1 is smaller than 160 GB.

What should you do?

- A. On the master database, execute a query against the sys.dm_pdw_nodes_os_performance_counters dynamic management view.

- B. From Azure Monitor in the Azure portal, execute a query against the logs of DW1.
- C. On DW1, execute a query against the sys.database_files dynamic management view.
- D. Execute a query against the logs of DW1 by using theGet-AzOperationalInsightSearchResult PowerShell cmdlet.

Answer: A

Explanation:

The following query returns the transaction log size on each distribution. If one of the log files is reaching 160 GB, you should consider scaling up your instance or limiting your transaction size.

```
-- Transaction log size SELECT
instance_name as distribution_db, cntr_value*1.0/1048576 as log_file_size_used_GB, pdw_node_id
FROM sys.dm_pdw_nodes_os_performance_counters WHERE
instance_name like 'Distribution_%'
AND counter_name = 'Log File(s) Used Size (KB)' References:
https://docs.microsoft.com/en-us/azure/sql-data-warehouse/sql-data-warehouse-manage-monitor
```

NEW QUESTION 77

- (Exam Topic 3)

You have an Azure Data Lake Storage Gen2 container that contains 100 TB of data.

You need to ensure that the data in the container is available for read workloads in a secondary region if an outage occurs in the primary region. The solution must minimize costs.

Which type of data redundancy should you use?

- A. zone-redundant storage (ZRS)
- B. read-access geo-redundant storage (RA-GRS)
- C. locally-redundant storage (LRS)
- D. geo-redundant storage (GRS)

Answer: C

NEW QUESTION 79

- (Exam Topic 3)

You are creating an Azure Data Factory data flow that will ingest data from a CSV file, cast columns to specified types of data, and insert the data into a table in an Azure Synapse Analytic dedicated SQL pool. The CSV file contains three columns named username, comment, and date.

The data flow already contains the following:

- A source transformation.
- A Derived Column transformation to set the appropriate types of data.
- A sink transformation to land the data in the pool.

You need to ensure that the data flow meets the following requirements:

- All valid rows must be written to the destination table.
- Truncation errors in the comment column must be avoided proactively.
- Any rows containing comment values that will cause truncation errors upon insert must be written to a file in blob storage.

Which two actions should you perform? Each correct answer presents part of the solution. NOTE: Each correct selection is worth one point.

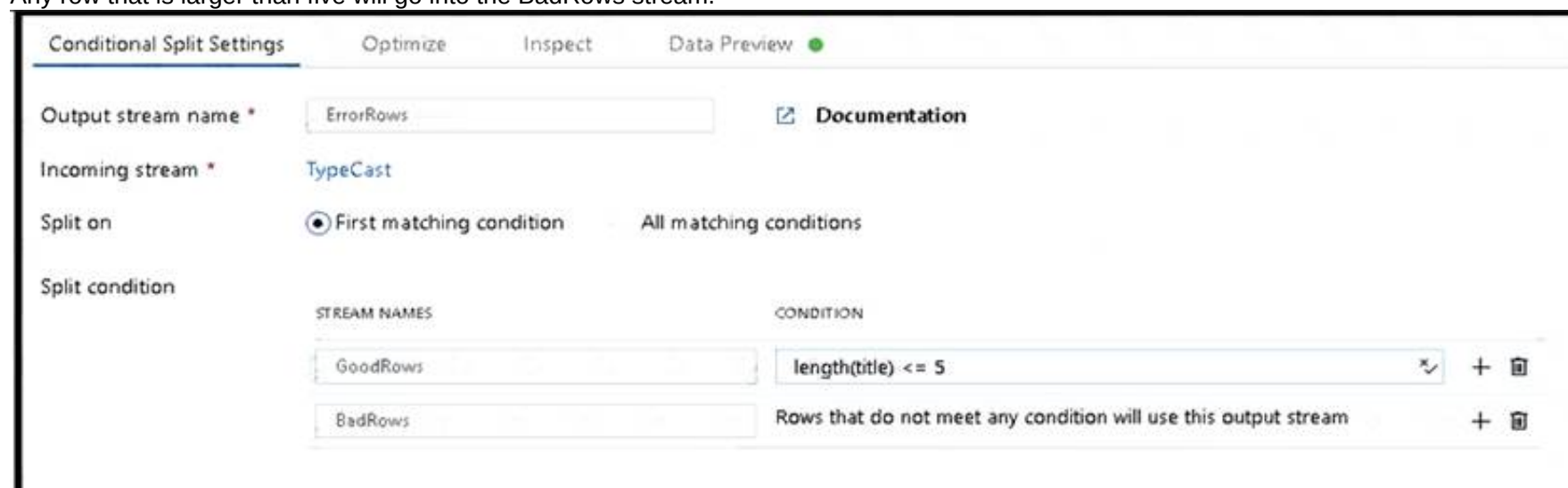
- A. To the data flow, add a sink transformation to write the rows to a file in blob storage.
- B. To the data flow, add a Conditional Split transformation to separate the rows that will cause truncation errors.
- C. To the data flow, add a filter transformation to filter out rows that will cause truncation errors.
- D. Add a select transformation to select only the rows that will cause truncation errors.

Answer: AB

Explanation:

B: Example:

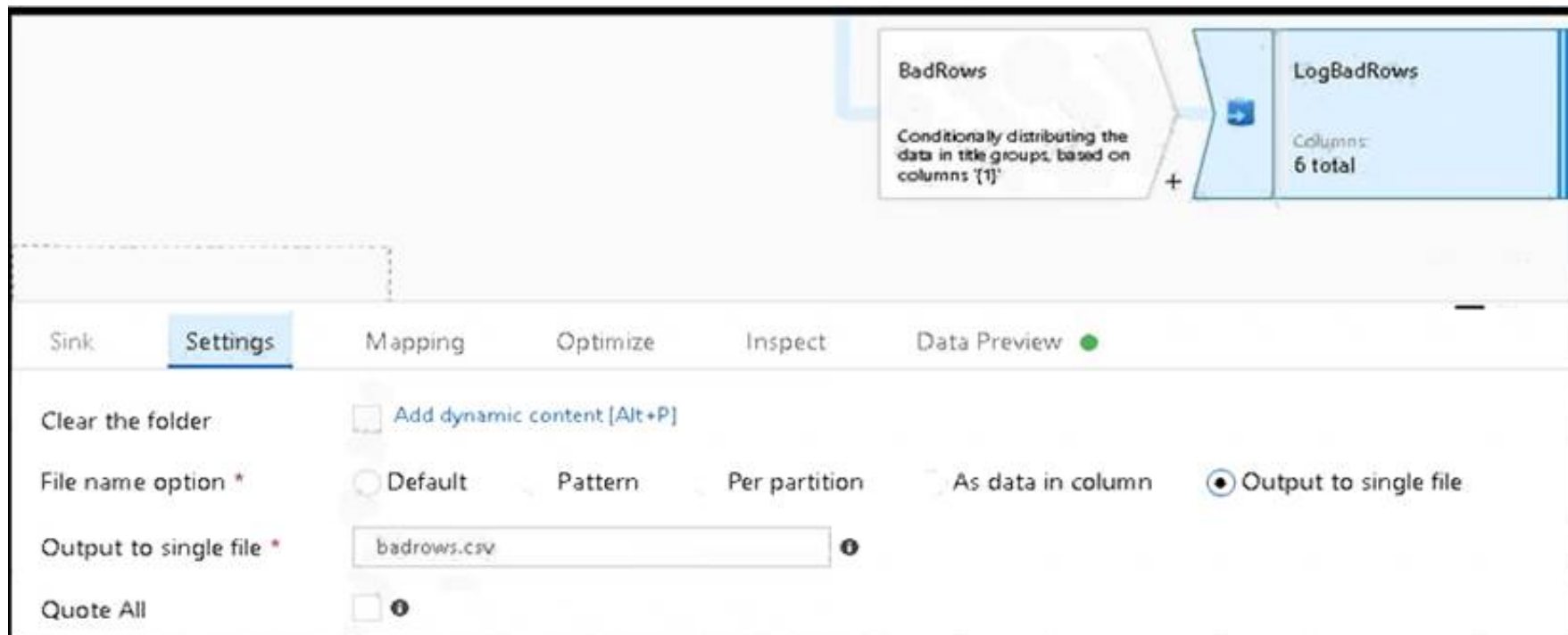
* 1. This conditional split transformation defines the maximum length of "title" to be five. Any row that is less than or equal to five will go into the GoodRows stream. Any row that is larger than five will go into the BadRows stream.



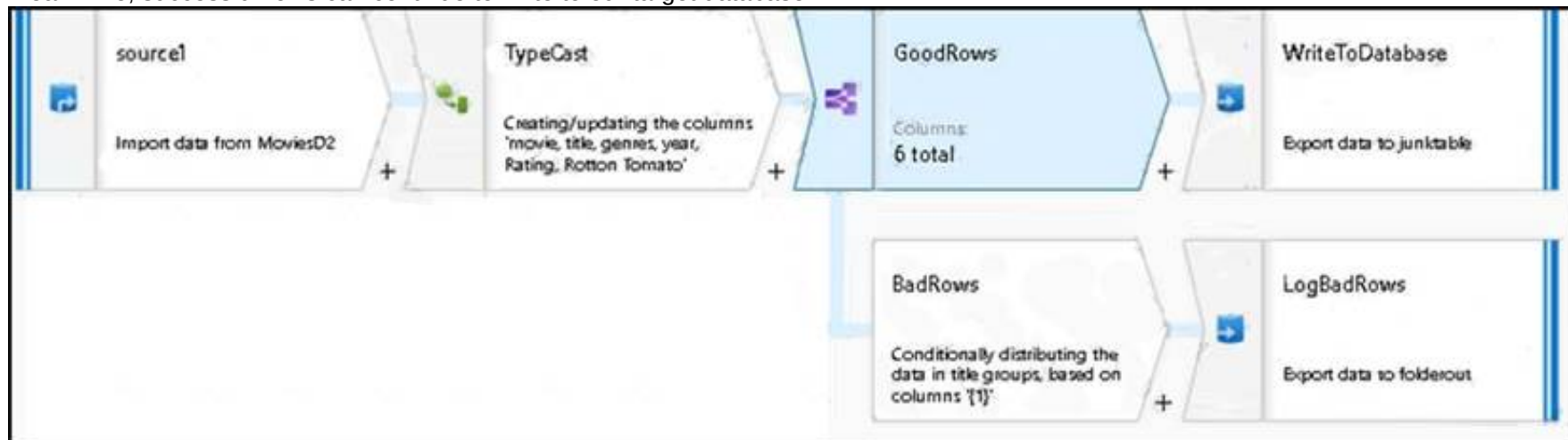
* 2. This conditional split transformation defines the maximum length of "title" to be five. Any row that is less than or equal to five will go into the GoodRows stream. Any row that is larger than five will go into the BadRows stream.

A:

* 3. Now we need to log the rows that failed. Add a sink transformation to the BadRows stream for logging. Here, we'll "auto-map" all of the fields so that we have logging of the complete transaction record. This is a text-delimited CSV file output to a single file in Blob Storage. We'll call the log file "badrows.csv".



* 4. The completed data flow is shown below. We are now able to split off error rows to avoid the SQL truncation errors and put those entries into a log file. Meanwhile, successful rows can continue to write to our target database.



Reference:

<https://docs.microsoft.com/en-us/azure/data-factory/how-to-data-flow-error-rows>

NEW QUESTION 81

- (Exam Topic 3)

You are designing an Azure Stream Analytics solution that receives instant messaging data from an Azure event hub. You need to ensure that the output from the Stream Analytics job counts the number of messages per time zone every 15 seconds.

How should you complete the Stream Analytics query? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

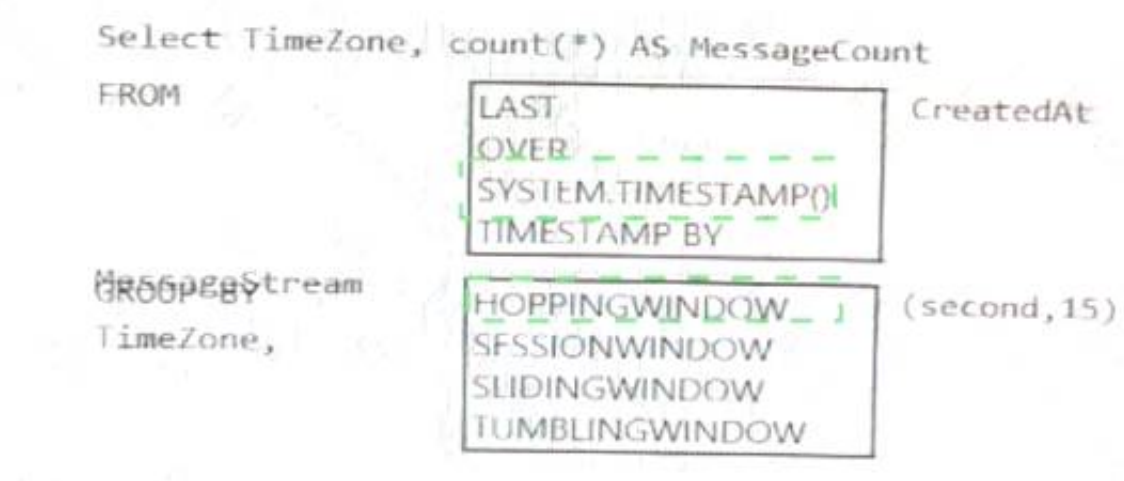
```
Select TimeZone, count(*) AS MessageCount
FROM
    MessageStream
GROUP BY
    TimeZone,
    LAST
    OVER
    SYSTEM.TIMESTAMP()
    TIMESTAMP BY
    HOPPINGWINDOW
    SESSIONWINDOW
    SLIDINGWINDOW
    TUMBLINGWINDOW
    (second, 15)
```

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

Answer Area



NEW QUESTION 85

- (Exam Topic 3)

You plan to ingest streaming social media data by using Azure Stream Analytics. The data will be stored in files in Azure Data Lake Storage, and then consumed by using Azure Databricks and PolyBase in Azure Synapse Analytics.

You need to recommend a Stream Analytics data output format to ensure that the queries from Databricks and PolyBase against the files encounter the fewest possible errors. The solution must ensure that the files can be queried quickly and that the data type information is retained.

What should you recommend?

- A. Parquet
- B. Avro
- C. CSV
- D. JSON

Answer: B

Explanation:

The Avro format is great for data and message preservation. Avro schema with its support for evolution is essential for making the data robust for streaming architectures like Kafka, and with the metadata that schema provides, you can reason on the data. Having a schema provides robustness in providing meta-data about the data stored in Avro records which are self-documenting the data. References: <http://cloudurable.com/blog/avro/index.html>

NEW QUESTION 88

- (Exam Topic 3)

Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.

After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.

You are designing an Azure Stream Analytics solution that will analyze Twitter data.

You need to count the tweets in each 10-second window. The solution must ensure that each tweet is counted only once.

Solution: You use a session window that uses a timeout size of 10 seconds. Does this meet the goal?

- A. Yes
- B. No

Answer: B

Explanation:

Instead use a tumbling window. Tumbling windows are a series of fixed-sized, non-overlapping and contiguous time intervals. Reference:

<https://docs.microsoft.com/en-us/stream-analytics-query/tumbling-window-azure-stream-analytics>

NEW QUESTION 92

- (Exam Topic 3)

You are designing an Azure Databricks interactive cluster. The cluster will be used infrequently and will be configured for auto-termination.

You need to ensure that the cluster configuration is retained indefinitely after the cluster is terminated. The solution must minimize costs.

What should you do?

- A. Clone the cluster after it is terminated.
- B. Terminate the cluster manually when processing completes.
- C. Create an Azure runbook that starts the cluster every 90 days.
- D. Pin the cluster.

Answer: D

Explanation:

To keep an interactive cluster configuration even after it has been terminated for more than 30 days, an administrator can pin a cluster to the cluster list.

References:

<https://docs.azuredatabricks.net/clusters/clusters-manage.html#automatic-termination>

NEW QUESTION 96

- (Exam Topic 3)

You have an Azure Synapse Analytics dedicated SQL pool that contains a table named Table1. You have files that are ingested and loaded into an Azure Data

Lake Storage Gen2 container named container1.

You plan to insert data from the files into Table1 and azure Data Lake Storage Gen2 container named container1.

You plan to insert data from the files into Table1 and transform the data. Each row of data in the files will produce one row in the serving layer of Table1.

You need to ensure that when the source data files are loaded to container1, the DateTime is stored as an additional column in Table1.

Solution: In an Azure Synapse Analytics pipeline, you use a Get Metadata activity that retrieves the DateTime of the files.

Does this meet the goal?

- A. Yes
- B. No

Answer: B

NEW QUESTION 99

- (Exam Topic 3)

You have an Azure Synapse Analytics dedicated SQL pool that contains a table named Contacts. Contacts contains a column named Phone.

You need to ensure that users in a specific role only see the last four digits of a phone number when querying the Phone column.

What should you include in the solution?

- A. a default value
- B. dynamic data masking
- C. row-level security (RLS)
- D. column encryption
- E. table partitions

Answer: C

NEW QUESTION 102

- (Exam Topic 3)

You have a table in an Azure Synapse Analytics dedicated SQL pool. The table was created by using the following Transact-SQL statement.

```
CREATE TABLE [dbo].[DimEmployee] (
    [EmployeeKey] [int] IDENTITY(1,1) NOT NULL,
    [EmployeeID] [int] NOT NULL,
    [FirstName] [varchar](100) NOT NULL,
    [LastName] [varchar](100) NOT NULL,
    [JobTitle] [varchar](100) NULL,
    [LastHireDate] [date] NULL,
    [StreetAddress] [varchar](500) NOT NULL,
    [City] [varchar](200) NOT NULL,
    [StateProvince] [varchar](50) NOT NULL,
    [Postalcode] [varchar](10) NOT NULL
)
```

You need to alter the table to meet the following requirements:

- Ensure that users can identify the current manager of employees.
- Support creating an employee reporting hierarchy for your entire company.
- Provide fast lookup of the managers' attributes such as name and job title.

Which column should you add to the table?

- A. [ManagerEmployeeID] [int] NULL
- B. [ManagerEmployeeID] [smallint] NULL
- C. [ManagerEmployeeKey] [int] NULL
- D. [ManagerName] [varchar](200) NULL

Answer: A

Explanation:

Use the same definition as the EmployeeID column. Reference:

<https://docs.microsoft.com/en-us/analysis-services/tabular-models/hierarchies-ssas-tabular>

NEW QUESTION 107

.....

Relate Links

100% Pass Your DP-203 Exam with ExamBible Prep Materials

<https://www.exambible.com/DP-203-exam/>

Contact us

We are proud of our high-quality customer service, which serves you around the clock 24/7.

Viste - <https://www.exambible.com/>